

MotionDNA: AI-Powered Markerless Biomechanical Analysis for Injury Risk Prediction

Project Report — Review 2 Draft

Course: Biomechanics (BBMD206L)

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Page 1 — Work Completed Since Review 1

Since Review 1, our team has progressed from preliminary literature review and initial architecture planning into a fully functional software prototype. The codebase is structured as a modular Python package (`motiondna/`) delivered through an intuitive Streamlit web dashboard (`app.py`). This operationalizes the complete end-to-end processing pipeline originally proposed:

`Video/CSV Input` $\rightarrow$ `Landmark Extraction` $\rightarrow$ `Knee Kinematics`
 $\rightarrow$ `Rule-Based Risk Screening` $\rightarrow$ `Interactive Dashboard & CSV Export`

Module-by-Module Summary of Completed Work

- `app.py` (**User Interface & Interactive Inspector**):

Serves as the primary entry point for end users. Beyond accepting CSV coordinate files or raw video uploads, running the analysis pipeline, and displaying overall risk metrics, `app.py` now includes an **Interactive Frame-by-Frame Inspector**. Users can scrub through the video timeline using a frame slider to inspect instantaneous joint parameters (flexion angles, asymmetry, velocity, and offsets) alongside a synchronized red cursor on the Plotly kinematic graph.

- `motiondna/pose.py` (**Pose Estimation & Overlay Generation**):

Uses Google's MediaPipe Pose model to detect lower-limb joint coordinates (hips, knees, and ankles) across video frames. During extraction, it scales raw relative coordinates using an estimated participant height to approximate metric distances. Crucially, `pose.py` now retains full-sequence, pose-annotated image frames with color-coded lower-limb skeleton overlays, allowing users to visually cross-verify tracking quality for every frame.

- `motiondna/kinematics.py` (**Calculation Core**):

The mathematical core of MotionDNA. It computes 3D vector-based knee flexion angles ($\theta_{\text{flexion}} = 180^\circ - \theta_{\text{included}}$), calculates numerical angular velocities ($\Delta \theta / \Delta t$), and measures horizontal knee-to-ankle offsets as a proxy for dynamic knee valgus. It also includes a synthetic squat and drop-jump generator for benchmarking while physical data collection is underway.

- `motiondna/risk.py` (**Rule-Based Screening Engine**):

Implements a transparent, fully explainable 0–100 risk scoring algorithm rather than an opaque black-box ML model. Risk points accumulate based on shallow squat/landing depth, bilateral flexion asymmetry ($>12^\circ$), excessive knee-ankle offset ($>4 \text{ cm}$), and high angular velocity ($>500^\circ/\text{s}$). Results map to clear risk bands (*Lower, Moderate, Elevated*) accompanied by plain-language clinical observations explaining *why* a score was given.

- `motiondna/__init__.py`:

Exposes key package functions (`analyze_landmarks`, `make_demo_trial`, `score_movement_risk`) for clean imports.

- `requirements.txt`, `packages.txt`, `runtime.txt`, `README.md`:

Documents project dependencies, setup commands, CSV schemas, and system-level Linux dependencies (`libgl1`, `libegl1`, `libgles2`, `libgl1-mesa-dev`) required for headless deployment on Streamlit Community Cloud.

[Space for additional notes / Reviewer Feedback comments]

This working prototype directly fulfills the two core objectives set out in Review 1: **Objective 1** (pose estimation pipeline) is realized in `pose.py`, while **Objective 2** (kinematic calculations, explainable risk scoring, and interactive frame inspection) is realized in `kinematics.py`, `risk.py`, and `app.py`.

The upcoming project phase will focus on collecting real participant movement trials (squats and drop jumps) to replace synthetic demo data and benchmarking our rule-based scoring against expert clinical assessments.

Draft Article

1. Introduction

Musculoskeletal injuries represent one of the most widespread and costly health challenges globally. The World Health Organization estimates that approximately **1.71 billion people worldwide** suffer from musculoskeletal conditions, making them the single largest contributor to disability worldwide, with low back pain leading disability in 160 countries. Economically, a 2021 global study calculated the welfare loss from musculoskeletal disorders at **over \$2.1 trillion USD** (1.41% of global GDP). In sports and athletics, lower-limb ligament injuries—particularly anterior cruciate ligament (ACL) tears—remain a severe concern, with an average incidence of roughly 1 in 3,500 in general athletic populations, and significantly higher rates among female field-sport athletes.

The core problem this project addresses is **access and objectivity**. The current gold standard for biomechanical injury-risk screening—marker-based 3D motion capture systems such as Vicon—is highly accurate but extremely expensive, hardware-heavy, and restricted to dedicated laboratory environments. Consequently, the vast majority of sports clubs, physiotherapy clinics, schools, and individual athletes cannot access objective screening. Instead, injury risk assessment typically falls on the visual judgment of a coach or physical therapist. While valuable, visual screening is inherently subjective, prone to human error, and suffers from high inter-rater variability.

This creates a critical gap: the populations without access to expensive biomechanical labs are often the ones who need early screening most. Without objective tools, subtle movement flaws go unnoticed until they lead to injury, return-to-sport decisions after rehabilitation remain subjective, and the gap widens between elite institutions and grassroots athletics.

MotionDNA was developed to bridge this exact gap. It is an AI-powered, markerless biomechanical screening tool that runs directly from smartphone or webcam video. It requires no physical body markers or specialized hardware, offering an interpretable risk score, detailed kinematic graphs, and a frame-by-frame breakdown through an accessible web dashboard.

[Space for Figure 1: MotionDNA High-Level System Overview]

2. Background

2.1 Literature Review

The technical foundation for markerless human pose estimation has advanced rapidly. Toshev and Szegedy's *DeepPose* (CVPR 2014) demonstrated that deep convolutional neural networks could directly regress joint coordinates from images, establishing the viability of deep learning over hand-crafted feature extractors. Chen and Ramanan (CVPR 2017) extended this by showing that 3D human pose could be approximated by lifting 2D joint

predictions via nearest-neighbor matching against a 3D pose library—proving that meaningful 3D spatial insights could be gained without multi-camera rigs or depth sensors.

Recent domain-specific literature highlights both the progress and remaining challenges in sports science. Meng et al. (2023) reviewed markerless motion analysis techniques in sports, noting a persistent accuracy gap relative to marker-based gold standards—a trade-off MotionDNA explicitly accepts in exchange for low cost and portability. Ru et al. (2025) surveyed deep-learning-based motion capture applications, confirming a steady rise in real-world adoption for injury prevention, rehabilitation tracking, and performance monitoring.

2.2 Prior Art and Its Limitations

Several existing markerless systems clarify the specific gap MotionDNA aims to fill:

- **Theia3D:** Uses synchronized multi-camera setups and deep learning to achieve high kinematic precision (reported sagittal plane RMSE of 3.2° to 15.7° against Vicon). However, it requires expensive camera arrays and dedicated processing hardware, reintroducing high cost and complexity.
- **OpenCap (Stanford University):** A free, smartphone-based multi-camera platform for gait and movement analysis. While closer to MotionDNA's goals, published validation studies note clinically meaningful errors in certain planes (e.g., sagittal hip-angle errors exceeding 10°), which can limit screening accuracy for ACL-related hip/trunk control.
- **DARI Motion, Sparta Science, & Kinatrax:** Commercial platforms widely used in professional sports for performance profiling, but priced strictly for enterprise/institutional budgets rather than individual clinicians, grassroots teams, or athletes.

The Gap MotionDNA Targets: None of these solutions combine (a) a single smartphone or webcam as the sole hardware requirement, (b) fully transparent, rule-based risk scoring, and (c) an interactive, zero-installation web dashboard featuring frame-by-frame inspection.

3. Methods

3.1 Technology Stack

- **Pose Estimation Engine:** Google MediaPipe Pose for tracking 3D hip, knee, and ankle keypoints from video frames.
- **Backend Processing:** Modular Python (`motiondna/`) separating pose extraction, kinematic calculations, and risk heuristics into clean, testable modules.
- **Frontend Dashboard:** Streamlit, enabling an interactive web interface with real-time Plotly charts and frame scrubbing.
- **Data Interoperability:** Support for raw video uploads (`.mp4`, `.mov`, `.avi`) and direct CSV landmark import/export.

3.2 System Pipeline & Workflow Logic

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$$\begin{aligned}
&\text{[Video / CSV Input]} \rightarrow \text{[pose.py] MediaPipe Keypoint Tracking \& Annotated Frame Capture} \\
&\rightarrow \text{[kinematics.py] Joint Angle, Velocity, \& Offset Calculations} \\
&\rightarrow \text{[risk.py] Transparent Heuristic Rule Scoring Engine} \\
&\rightarrow \text{[app.py] Dashboard: Metrics, Line Plots, Frame Inspector, CSV Export}
\end{aligned}$$

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3.3 Risk Scoring Logic & Interactive Frame Inspection

The risk engine uses transparent rules so every point added to the composite score ($0 \text{--} 100$) traces directly back to an observable biomechanical factor:

1. **Flexion Depth:** Points added for shallow squatting ($< 70^\circ$) or landing ($< 35^\circ$).
2. **Side-to-Side Symmetry:** Points added if left/right knee flexion difference exceeds 12° .
3. **Knee–Ankle Alignment Proxy:** Points added if horizontal knee-to-ankle medial offset exceeds 4 cm (dynamic valgus proxy).
4. **Angular Velocity:** Points added if peak landing velocity exceeds $500^\circ/\text{s}$.

To complement the aggregate score, the dashboard includes a **Frame-by-Frame Inspector**. Users can drag a timeline slider to freeze the movement at critical moments (e.g., maximum squat depth or initial landing contact). As the slider moves:

- The corresponding video frame displays with the MediaPipe lower-limb skeleton overlaid.
- Instantaneous joint angles, asymmetry, velocity, and offsets update in real-time.
- A vertical red indicator line moves across the Plotly time-series chart, visually linking the image frame to its exact spot on the kinematic curve.

[Space for Figure 2: Interactive Frame Inspector & Synchronized Timeline Chart]

4. Results (Expected)

The pipeline has been validated end-to-end using synthetic squat and drop-jump trials generated in `kinematics.py`, confirming internal mathematical consistency (e.g., higher asymmetry directly scales the composite risk score). As real participant data collection gets underway in the coming weeks, the following outcomes are expected:

- **Visual Validation:** Knee flexion angles and velocities that track smoothly against video overlays.
- **Risk Stratification:** Clear numerical separation between controlled, well-aligned movement patterns and those exhibiting visible valgus collapse or asymmetry.
- **Error Quantification:** Quantified accuracy metrics (e.g., RMSE) benchmarking MotionDNA's single-camera output against reference video analysis.
- **Usability & Interpretability:** Positive feedback from coaches and student-athletes on the clarity of the frame-by-frame inspector and plain-language risk findings.

5. Conclusion

MotionDNA demonstrates that a single-camera, markerless system can deliver screening-grade biomechanical insights without sacrificing transparency or requiring expensive lab infrastructure. By combining real-time joint kinematic tracking with explainable risk rules and an intuitive frame-by-frame scrubbing interface, MotionDNA empowers coaches, clinicians, and athletes to identify risky movement patterns early and objectively.

6. Future Recommendations

1. **Longitudinal Progress Tracking:** Store session data over time to monitor rehabilitation progress and movement recovery.
2. **Expanded Movement Library:** Extend rules to sport-specific movements like cutting, single-leg hopping, and sprint decelerations.
3. **Hybrid ML Refinement:** Layer an explainable machine learning model on top of the rule engine once larger datasets are collected.
4. **Mobile Deployment:** Package the web application into a mobile-first PWA for easy field use on tablets and smartphones.

UN Sustainable Development Goals Alignment:

- **SDG 3 (Good Health and Well-Being):** Reduces preventable joint injuries through early, accessible screening.
- **SDG 10 (Reduced Inequalities):** Democratizes access to biomechanical analysis for under-resourced sports programs and clinics.

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